
Relations

Comprehensive Lecture Notes

COT3100C — Honors Introduction to Discrete Structures

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Color guide

Blue	formal definitions	Gold	worked examples
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1 Cartesian Products and Relations

1.1 Cartesian products

For sets A and B , the *Cartesian product* is the set of all ordered pairs whose first coordinate comes from A and whose second coordinate comes from B :

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Order matters: in general, $(a, b) \neq (b, a)$ and $A \times B \neq B \times A$. If $|A| = m$ and $|B| = n$ are finite, then $|A \times B| = mn$.

Example: forming a Cartesian product

If $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$, then

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3), (c, 1), (c, 2), (c, 3)\}.$$

1.2 Binary and n -ary relations

Binary relation

A *binary relation from A to B* is any subset $R \subseteq A \times B$. If $(x, y) \in R$, we write either xRy or “ x is R -related to y .” If $(x, y) \notin R$, then xRy is false.

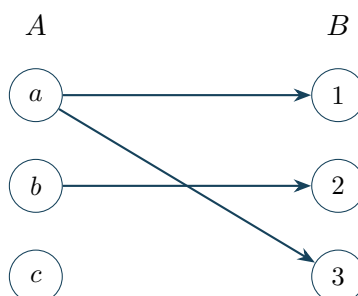
In plain English

Read $R \subseteq A \times B$ as: “ R is a selection of the possible ordered pairs from A and B .” The Cartesian product lists every possible connection; the relation keeps only the connections that satisfy the rule we care about.

Thus a relation makes a yes/no statement about each possible ordered pair. For example, the relation

$$R = \{(a, 1), (a, 3), (b, 2)\} \subseteq A \times B$$

contains three of the nine possible pairs. The arrow diagram below records exactly those three pairs.



A relation *on* A means $R \subseteq A \times A$. Familiar examples include $=$, $<$, $\leq$, divisibility, and congruence modulo m .

The same idea extends to more coordinates. An *n -ary relation* on $A_1, \dots, A_n$ is a subset of $A_1 \times \dots \times A_n$; its *degree* is n . A database table can be viewed as an n -ary relation: each row is one n -tuple in the relation.

1.3 Functions are special relations

Function as a relation

A function $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ satisfying

$$\forall x \in A \exists! y \in B \text{ such that } (x, y) \in f.$$

The symbol $\exists!$ means “there exists exactly one.” Therefore every input in A must occur, and it must be paired with only one output.

Reading the quantified statement

The formula says: “For every input x in A , there is one and only one output y in B paired with it.” Think *one output per input*: no missing inputs and no input that branches to two outputs.

The relation in the arrow diagram above is *not* a function $A \rightarrow B$: the input a has two outputs, while c has no output. Multiple inputs may share one output; that does not violate the definition of a function.

2 Representing a Relation

A finite relation can be displayed as a set of ordered pairs, an arrow diagram, a directed graph (digraph), or a zero–one matrix. These are different encodings of the same information.

2.1 Zero–one matrices

Let $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_n\}$. The matrix of $R \subseteq A \times B$ is $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1, & (a_i, b_j) \in R, \\ 0, & (a_i, b_j) \notin R. \end{cases}$$

Rows correspond to possible first coordinates, and columns correspond to possible second coordinates. For the preceding relation, using the listed orders of A and B ,

$$M_R = \begin{array}{c|ccc} & 1 & 2 & 3 \\ \hline a & 1 & 0 & 1 \\ b & 0 & 1 & 0 \\ c & 0 & 0 & 0 \end{array}.$$

A relation $A \rightarrow B$ is a function exactly when every row of its matrix contains exactly one 1.

Matrix reading rule

To decide whether (a_i, b_j) belongs to the relation, go to row a_i and column b_j . A 1 means “yes, this pair is in R ”; a 0 means “no.”

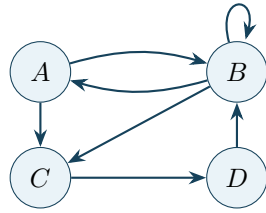
2.2 Directed graphs and adjacency matrices

For a relation on one finite set A , draw one vertex for each element of A and an arrow $x \rightarrow y$ exactly when xRy . A pair (x, x) appears as a loop at x .

For example, let $A = \{A, B, C, D\}$ and

$$R = \{(A, B), (A, C), (B, A), (B, B), (B, C), (C, D), (D, B)\}.$$

Then the digraph and its adjacency matrix are:



$$M_R = \begin{array}{c|cccc} & A & B & C & D \\ \hline A & 0 & 1 & 1 & 0 \\ B & 1 & 1 & 1 & 0 \\ C & 0 & 0 & 0 & 1 \\ D & 0 & 1 & 0 & 0 \end{array}$$

For an undirected graph, each ordinary edge $\{x, y\}$ represents both (x, y) and (y, x) , so its adjacency matrix is symmetric. A directed graph need not have a symmetric matrix.

3 Properties of Relations on a Set

In this section $R \subseteq A \times A$.

Five common properties

- Reflexive:** $\forall x \in A, xRx$.
- Irreflexive:** $\forall x \in A, \neg(xRx)$.
- Symmetric:** $\forall x, y \in A, xRy \Rightarrow yRx$.
- Antisymmetric:** $\forall x, y \in A, (xRy \wedge yRx) \Rightarrow x = y$.
- Transitive:** $\forall x, y, z \in A, (xRy \wedge yRz) \Rightarrow xRz$.

The five properties as questions

- **Reflexive:** Does every element relate to itself?
- **Irreflexive:** Does no element relate to itself?
- **Symmetric:** Whenever a connection goes one way, must it also go back?
- **Antisymmetric:** Can a two-way connection occur only when the two endpoints are actually the same element?
- **Transitive:** If x reaches y and y reaches z , must x also reach z directly?

Two distinctions that prevent common mistakes

“Not reflexive” is not the same as “irreflexive.” A relation is not reflexive when even one required loop is missing; it is irreflexive only when *every* loop is absent. Likewise, symmetric and antisymmetric are not logical opposites. Equality is both symmetric and antisymmetric, while some relations are neither.

3.1 Reading the properties from a graph or matrix

Property	Digraph test	Matrix test
Reflexive	Every vertex has a loop.	Every diagonal entry is 1.
Irreflexive	No vertex has a loop.	Every diagonal entry is 0.
Symmetric	Every arrow $x \rightarrow y$ has the reverse arrow $y \rightarrow x$.	$M_R = M_R^T$.
Antisymmetric	No two distinct vertices have arrows in both directions.	If $i \neq j$ and $m_{ij} = 1$, then $m_{ji} = 0$.
Transitive	Every path $x \rightarrow y \rightarrow z$ has the shortcut $x \rightarrow z$.	With Boolean multiplication, $M_R^2 \leq M_R$ entrywise.

In Boolean matrix multiplication, multiplication is replaced by AND and addition by OR. Thus $(M_R^2)_{ik} = 1$ precisely when there is a two-edge path from a_i to a_k .

Why Boolean squaring tests transitivity

M_R^2 finds all destinations reachable in exactly two relation steps. Transitivity requires every such two-step destination to already be connected by a one-step arrow in M_R . That is what the entrywise condition $M_R^2 \leq M_R$ says.

3.2 Examples from the lecture

The following table gathers the relations compared in class. Each is a relation on the indicated standard number system.

Relation	Reflexive	Symmetric	Transitive
$x = y$ on any set	Yes	Yes	Yes
$x \leq y$ on $\mathbb{R}$	Yes	No	Yes
$x < y$ on $\mathbb{R}$	No (irreflexive)	No	Yes
$ x - y \leq 1$ on $\mathbb{Z}$	Yes	Yes	No
$x - y$ is even on $\mathbb{Z}$	Yes	Yes	Yes

For the fourth row, transitivity fails because $0R1$ and $1R2$, but $0 \not R2$. For the fifth row, if $x - y = 2k$ and $y - z = 2\ell$, then $x - z = 2(k + \ell)$, so the relation is transitive.

3.3 Divisibility

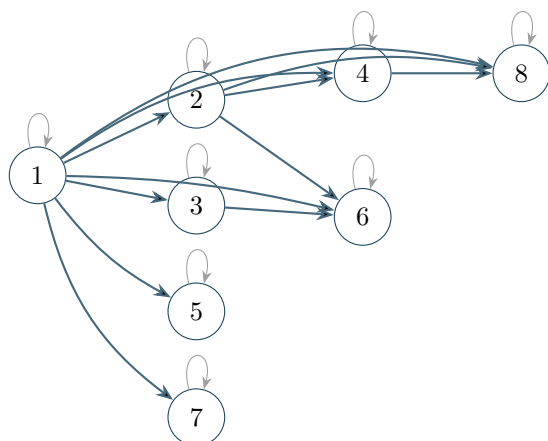
On the positive integers, define xRy when $x \mid y$. For the finite set $D = \{1, 2, 3, 4, 5, 6, 7, 8\}$, the full relation is

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), \\ (2, 2), (2, 4), (2, 6), (2, 8), (3, 3), (3, 6), (4, 4), (4, 8), \\ (5, 5), (6, 6), (7, 7), (8, 8)\}.$$

Reading divisibility notation

The statement $x \mid y$ is read “ x divides y .” It means that $y = xk$ for some integer k ; in other words, dividing y by x leaves no remainder. For example, $2 \mid 8$ because $8 = 2 \cdot 4$, but $3 \nmid 8$.

The loops show reflexivity. Divisibility is not symmetric: $2 \mid 4$, but $4 \nmid 2$. It is antisymmetric on positive integers, because $x \mid y$ and $y \mid x$ force $x = y$. It is transitive because $x \mid y$ and $y \mid z$ imply $y = xk$ and $z = y\ell$, hence $z = x(k\ell)$ and $x \mid z$.



An arrow $x \rightarrow y$ means $x \mid y$.
Loops and transitive arrows are shown.

Partial order

A relation is a *partial order* when it is reflexive, antisymmetric, and transitive. A set equipped with a partial order is a *partially ordered set* (poset). Divisibility on positive integers is a partial order. A partial order is *total* if every pair is comparable: for all x, y , either xRy or yRx . The usual $\leq$ is total; divisibility is not, since, for example, neither $2 \mid 3$ nor $3 \mid 2$.

In plain English

A partial order is a consistent way to arrange or compare objects, but it does not promise that every two objects can be compared. Divisibility orders numbers by “fits evenly into,” so 2 is below 8, while 2 and 3 are incomparable.

4 Equivalence Relations

Equivalence relation

A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive. It groups elements that are considered equivalent according to a chosen criterion.

In plain English

An equivalence relation is a valid “same group as” rule. Reflexivity puts every element in a group with itself, symmetry makes group membership mutual, and transitivity prevents overlapping groups from disagreeing about membership.

Equality is an equivalence relation. The relation “has the same parity as” on $\mathbb{Z}$ is another: it separates the integers into evens and odds. In general, any function $f : A \rightarrow B$ induces an equivalence relation on A by

$$x \sim y \iff f(x) = f(y).$$

For example, people can be related when they have the same age, or vectors can be related when they have the same length.

4.1 Congruence modulo 3

Define a relation on $\mathbb{Z}$ by

$$x \sim y \iff x \equiv y \pmod{3} \iff 3 \mid (x - y).$$

Reading the congruence equation

The statement $x \equiv y \pmod{3}$ says: “ x and y leave the same remainder when divided by 3.” Equivalently, their difference $x - y$ is a multiple of 3. For example, $8 \equiv 2 \pmod{3}$ because $8 - 2 = 6 = 3 \cdot 2$.

This is an equivalence relation:

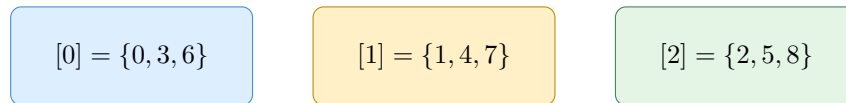
- **Reflexive:** $x - x = 0 = 3 \cdot 0$, so $3 \mid (x - x)$.
- **Symmetric:** if $x - y = 3k$, then $y - x = 3(-k)$.
- **Transitive:** if $x - y = 3k$ and $y - z = 3\ell$, then $x - z = (x - y) + (y - z) = 3(k + \ell)$.

Every integer has remainder 0, 1, or 2 upon division by 3. Therefore the three equivalence classes are

$$\begin{aligned} [0] &= \{3k : k \in \mathbb{Z}\}, \\ [1] &= \{3k + 1 : k \in \mathbb{Z}\}, \\ [2] &= \{3k + 2 : k \in \mathbb{Z}\}. \end{aligned}$$

On the finite sample $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, these become $\{0, 3, 6\}$, $\{1, 4, 7\}$, and $\{2, 5, 8\}$.

Three disconnected relation components



Within each box every ordered pair is related;
no ordered pair connects two different boxes.

If rows and columns are ordered by class as 0, 3, 6, 1, 4, 7, 2, 5, 8, the relation matrix has block form

$$M_{\sim} = \begin{pmatrix} J_3 & 0 & 0 \\ 0 & J_3 & 0 \\ 0 & 0 & J_3 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The all-one blocks record that every two elements in the same class are related.

4.2 Equivalence classes

Equivalence class and quotient set

If $\sim$ is an equivalence relation on A , the equivalence class of $x \in A$ is

$$[x] = \{a \in A : a \sim x\}.$$

The set of all distinct equivalence classes is the quotient

$$A/\sim = \{[x] : x \in A\}.$$

In plain English

$[x]$ is the entire group containing x —all elements equivalent to x . The quotient $A/\sim$ is not a set of individual elements; it is the set of all distinct groups. For modulo 3, there are exactly three groups, one for each possible remainder.

Reflexivity guarantees $x \in [x]$, so every class is nonempty. More strongly, for any $x, y \in A$, the following statements are equivalent:

$$x \sim y, \quad [x] \cap [y] \neq \emptyset, \quad [x] = [y].$$

Consequently, two equivalence classes are either identical or disjoint.

Why intersecting classes must be equal

Suppose $z \in [x] \cap [y]$. Then $z \sim x$ and $z \sim y$. Symmetry gives $x \sim z$, so transitivity gives $x \sim y$. If $a \in [x]$, then $a \sim x$ and $x \sim y$, hence $a \sim y$ and $a \in [y]$. Thus $[x] \subseteq [y]$; the reverse inclusion follows symmetrically, so $[x] = [y]$.

For congruence modulo 3,

$$\mathbb{Z}/\sim = \{[0], [1], [2]\}.$$

Different representatives may name the same class: for example, $[4] = [1]$ and $[-1] = [2]$ modulo 3.

5 Equivalence Relations and Partitions

Partition

A *partition* of a set A is a collection of nonempty subsets $A_1, \dots, A_k$ (or an indexed family in the infinite case) such that

$$A = \bigcup_i A_i \quad \text{and} \quad A_i \cap A_j = \emptyset \quad \text{whenever } i \neq j.$$

The subsets A_i are called blocks or parts.

In plain English

A partition sorts every element into exactly one nonempty bucket: the buckets cover the whole set, and no two buckets overlap. An equivalence relation tells us when two elements belong in the same bucket; the equivalence classes are the buckets.

Every equivalence relation on A produces a partition of A : its blocks are the distinct equivalence classes. Conversely, every partition produces an equivalence relation by declaring

$$x \sim y \iff x \text{ and } y \text{ lie in the same block.}$$

This relation is reflexive because every element lies in a block, symmetric because “being in the same block” is symmetric, and transitive because if x, y share a block and y, z share a block, uniqueness of the block containing y forces all three elements into that same block.

Thus equivalence relations and partitions are two descriptions of the same structure:

$$\boxed{\text{equivalence relation on } A} \iff \boxed{\text{partition of } A}.$$

6 Practice Questions

1. Let $A = \{a, b, c\}$, $B = \{1, 2, 3\}$, and

$$R = \{(a, 1), (a, 3), (b, 2)\}.$$

Write the zero-one matrix of R using the displayed orders of A and B . Is R a function from A to B ? Explain.

2. On $D = \{1, 2, 3, 4, 6, 8\}$, define xRy exactly when $x \mid y$. Determine whether R is reflexive, symmetric, antisymmetric, and transitive. Is it an equivalence relation? Is it a partial order?
3. On $\mathbb{Z}$, define xRy when $|x - y| \leq 1$. Determine whether R is reflexive, symmetric, and transitive. For every property that fails, give a counterexample.
4. On $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, define $x \sim y$ when $x \equiv y \pmod{3}$. Find all distinct equivalence classes and the quotient $A/\sim$. If the matrix rows and columns are ordered 0, 3, 6, 1, 4, 7, 2, 5, 8, describe the block structure of the matrix.

5. Let

$$\mathcal{P} = \{\{1, 4\}, \{2, 5, 8\}, \{3, 6, 7\}\}$$

be a partition of $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Define xRy when x and y belong to the same block of $\mathcal{P}$.

- (a) Prove that R is an equivalence relation.
- (b) Find $[4]$, $[5]$, and A/R .
- (c) Is $(1, 2)$ in R ? Is $(5, 8)$ in R ?

7 Answer Key

1. The matrix is

$$M_R = \begin{array}{c|ccc} & 1 & 2 & 3 \\ \hline a & 1 & 0 & 1 \\ b & 0 & 1 & 0 \\ c & 0 & 0 & 0 \end{array}.$$

It is not a function $A \rightarrow B$. Row a has two 1s, so a has two outputs; row c has no 1, so c has no output. A function matrix must have exactly one 1 in every row.

2. The relation is reflexive because $x \mid x$ for every $x \in D$. It is not symmetric: $2 \mid 4$, but $4 \nmid 2$. It is antisymmetric because for positive integers $x \mid y$ and $y \mid x$ imply $x = y$. It is transitive because $x \mid y$ and $y \mid z$ imply $x \mid z$. Therefore it is not an equivalence relation (symmetry fails), but it is a partial order.
3. It is reflexive because $|x - x| = 0 \leq 1$. It is symmetric because $|x - y| = |y - x|$. It is not transitive: $0R1$ and $1R2$, but $|0 - 2| = 2 > 1$, so $0 \nmid R2$.
4. The distinct classes are

$$[0] = \{0, 3, 6\}, \quad [1] = \{1, 4, 7\}, \quad [2] = \{2, 5, 8\}.$$

Hence

$$A/\sim = \{\{0, 3, 6\}, \{1, 4, 7\}, \{2, 5, 8\}\}.$$

With the specified order, the matrix is block diagonal with three 3×3 all-one blocks and zeros elsewhere:

$$M_\sim = \text{diag}(J_3, J_3, J_3).$$

5. (a) Every element lies in the same block as itself, so R is reflexive. If x and y are in the same block, then y and x are in the same block, so R is symmetric. If x and y are in the same block and y and z are in the same block, the unique block containing y contains all three; hence x and z are in the same block. Thus R is transitive. Therefore R is an equivalence relation.
- (b) $[4] = \{1, 4\}$, $[5] = \{2, 5, 8\}$, and

$$A/R = \{\{1, 4\}, \{2, 5, 8\}, \{3, 6, 7\}\}.$$

- (c) $(1, 2) \notin R$ because 1 and 2 are in different blocks. $(5, 8) \in R$ because 5 and 8 are in the same block.

Sources Consulted

- Patrick Keef and David Guichard, *An Introduction to Higher Mathematics*, Chapter 5 website: https://www.whitman.edu/mathematics/higher_math_online/chapter05.html
- Section 5.1, “Equivalence Relations”: https://www.whitman.edu/mathematics/higher_math_online/section05.01.html
- Section 5.2, “Factoring Functions”: https://www.whitman.edu/mathematics/higher_math_online/section05.02.html
- Section 5.3, “Ordered Sets”: https://www.whitman.edu/mathematics/higher_math_online/section05.03.html